

Winning the space race with data science

Historical Falcon 9 first-stage landing analysis | Applied Data Science Capstone

IBM Skills Network historical course datasets, analyzed with Python, SQL, Folium, Plotly and classification models.

90 launch records in the modeling cohort; data dated 2010 to 2020.

This report presents executed results. Its figures do not describe current launch performance.

Executive summary

Method and result in one view

Collected archived IBM course outputs for launch analysis, SQL, mapping and dashboard work. Cleaned the cohort, queried launch patterns and built interactive visuals.

60 of 90 historical modeling rows have a successful landing label (66.7%).

Four classifiers were tuned with five-fold cross-validation on 72 training rows. The best cross-validation score was 87.4%; held-out accuracy was 77.8% on 18 rows.

The selected model identified all 12 successful landings in the test set but misclassified 4 of 6 unsuccessful ones.

Introduction and business question

Can launch features anticipate first-stage landing success?

A successful first-stage landing supports rocket reuse. For a launch with known mission and booster features, predict Class = 1 for landing success.

The analytical unit is a Falcon 9 launch record. This study examines historical association and predictive performance; it does not estimate cost savings or causal effects.

Data window: 4 June 2010 through 5 November 2020 in the 90-row modeling cohort.

Data collection: SpaceX API

Archived IBM output from the course API workflow

Course collection sequence: request past launches from the SpaceX v4 API; look up related rocket, launchpad, payload and core records; filter Falcon 9; assemble launch features.

For this reproducible report, dataset_part_2.csv is the archived IBM Skills Network output from that workflow. It has 90 rows and 18 original columns.

The live API returned HTTP 525 during this revision. The analysis uses the archived course dataset and records its direct source URL in the README.

Data collection: web scraping

Historical launch table and SQL source

The course scraping workflow extracts Falcon 9 launch-table rows, normalizes column names, removes citation artifacts and parses payload mass and landing outcome.

The IBM Skills Network SpaceX.csv historical table supplies 101 records for SQLite analysis. The course datasets for mapping and dashboard each contain 56 records.

These datasets are separate historical extracts. Their counts and field names differ, so SQL, map, dashboard and model denominators are stated on each result slide.

Data wrangling methodology

Consistent outcomes and audit checks

Parsed Date to Year, retained the IBM binary Class target and validated its observed values. A value of 1 denotes a successful first-stage landing.

Checked nulls and duplicates. The modeling file has 26 missing LandingPad entries; that field was excluded from classification. Predictor columns used in the model have no missing entries.

One-hot encoded Orbit and LaunchSite. Standardization for logistic regression, SVM and KNN occurs inside training-fold pipelines.

EDA and visualization methodology

The question answered by each chart

Visual	Purpose	Denominator
Site bar	Compare landing rates by site	90
Orbit bar	Compare mission profiles	90
Year line	Inspect historical time pattern	90
Payload scatter	Inspect mass and outcome	90
Map	Locate recorded launch sites	56
Dashboard	Filter payload and success by site	56

EDA with SQL methodology

Queries executed in SQLite against SPACEXTBL

COUNT(*) GROUP BY Launch_Site identifies sites and ranks launch volume.

MIN, MAX and AVG summarize payload mass. CASE WHEN Landing_Outcome LIKE Success% counts successful landing records.

GROUP BY Orbit ranks orbit frequencies. SUBSTR(Date,1,4) groups launches and successful landings by year.

The SQL table has 101 rows and a different date and outcome schema from the 90-row modeling cohort.

Interactive visual analytics

Map and dashboard methods

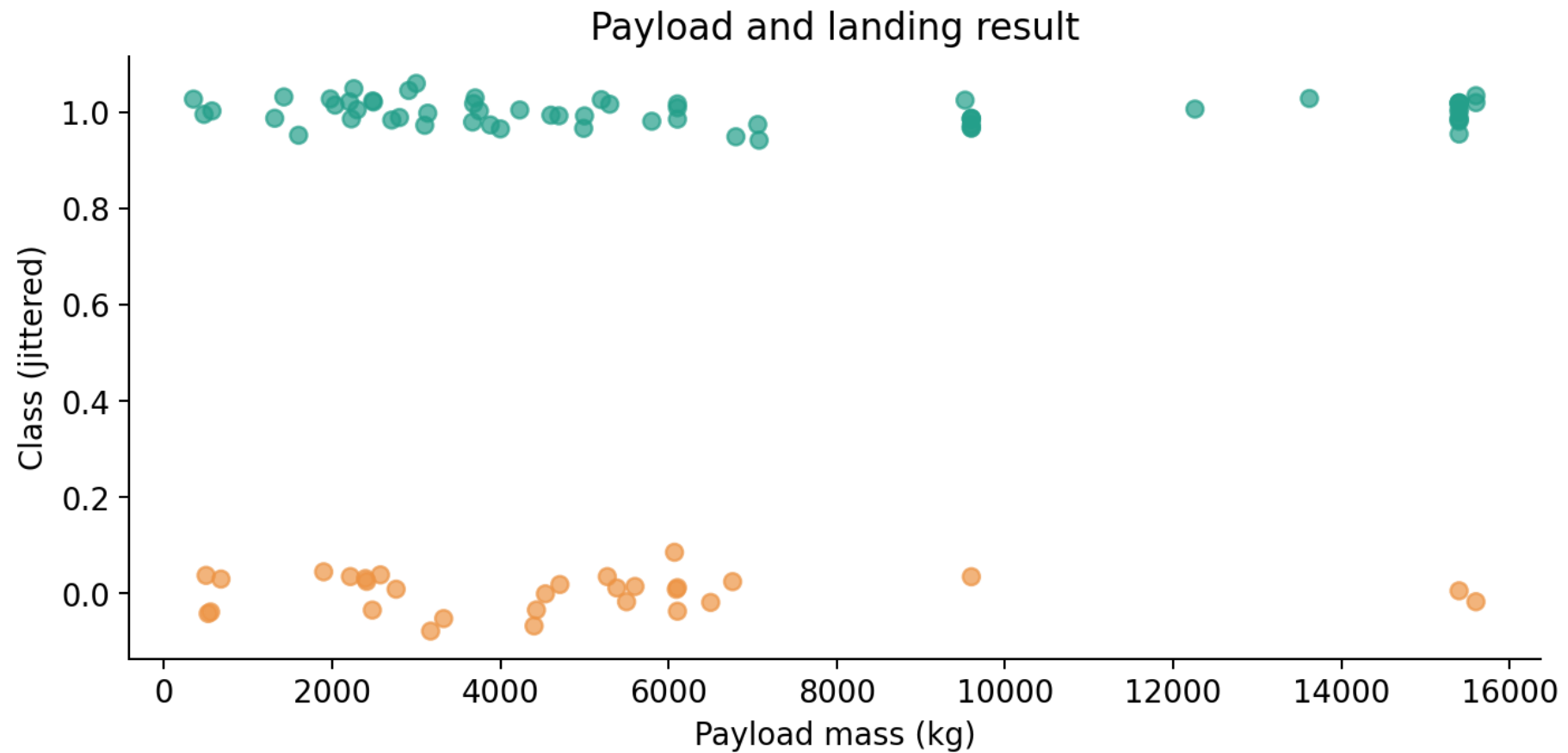
Folium: place one marker at each recorded launch site; tooltip reports site, row count and success fraction. A 5 km circle around each marker shows local proximity.

Plotly: a success-by-site pie chart and payload-mass versus landing-outcome scatter. Hover and site color reveal individual records.

The accompanying `folium_map.html` and `plotly_dashboard.html` open in a browser. The visual cohort has 56 historical records.

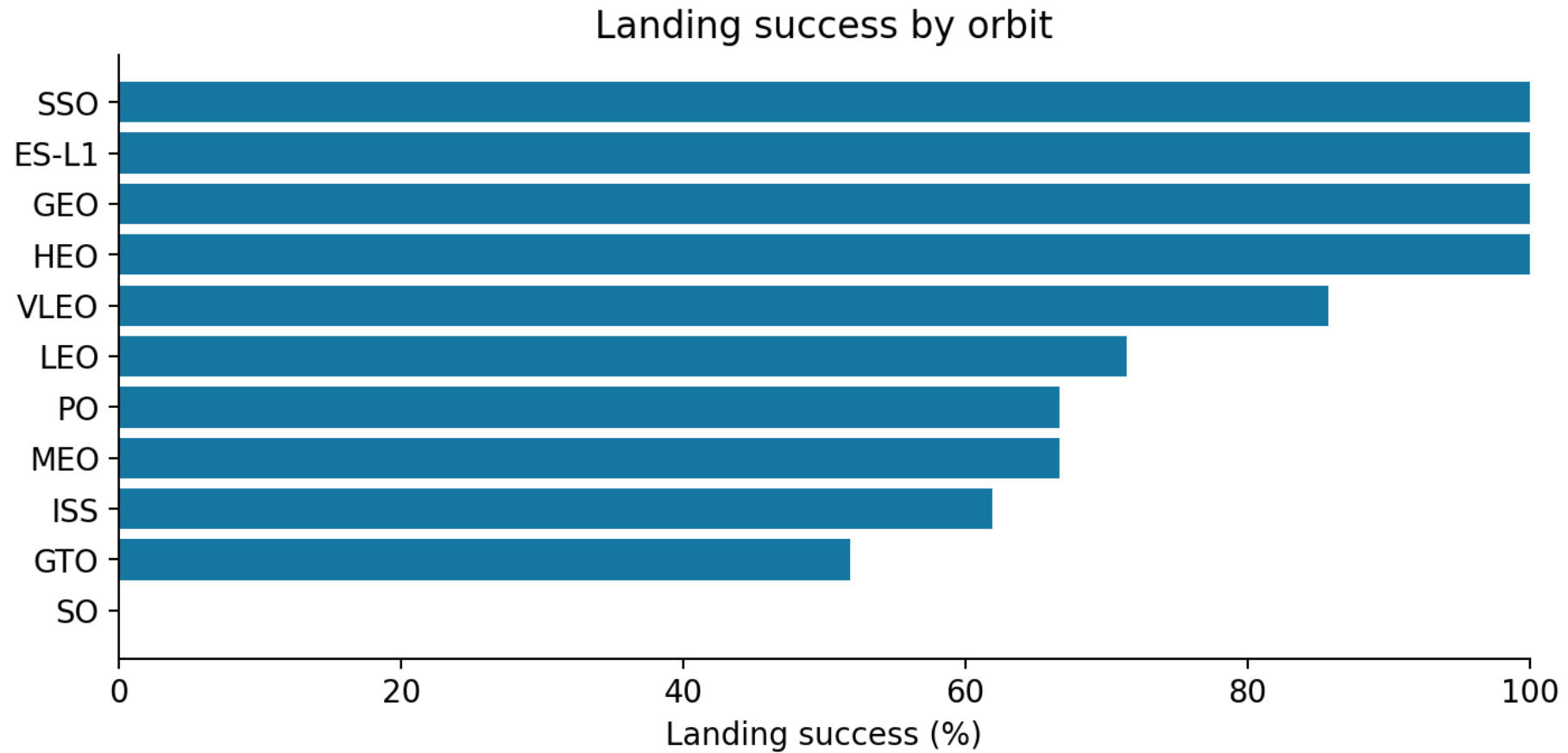
EDA: payload and landing

Lower and higher payloads occur in both outcome groups



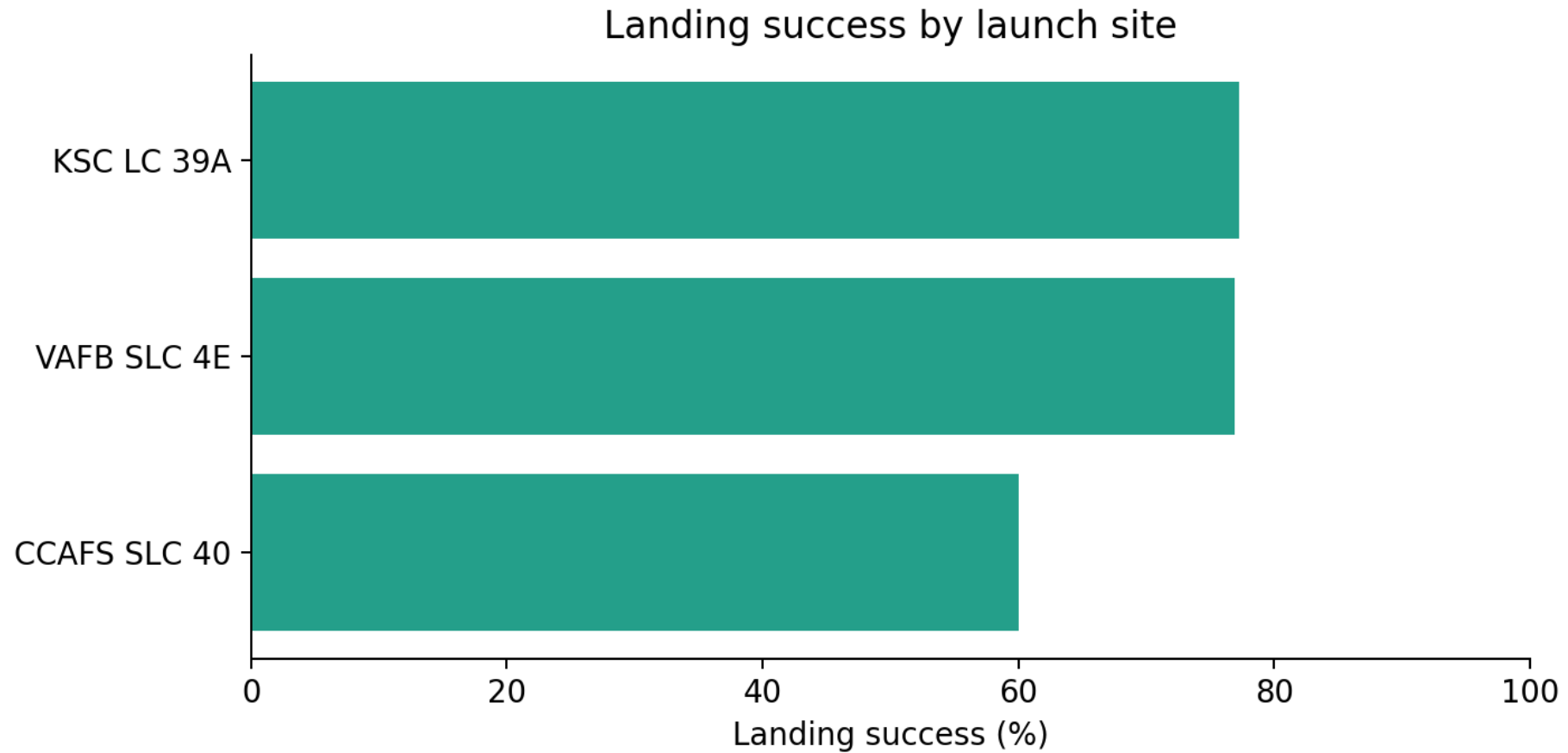
EDA: landing rate by orbit

Single-record orbit categories can show extreme percentages



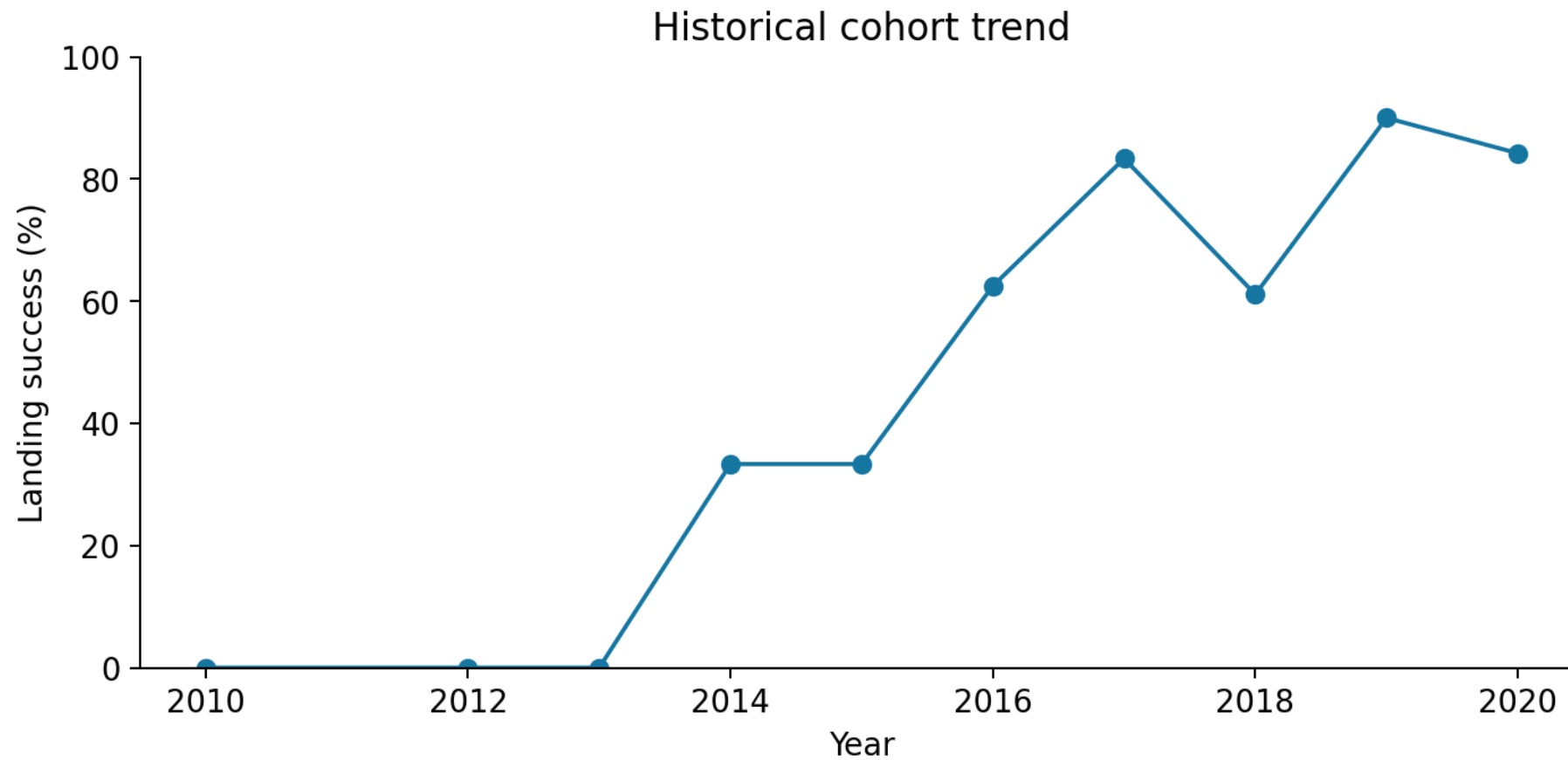
EDA: landing rate by site

Historical 90-row model cohort



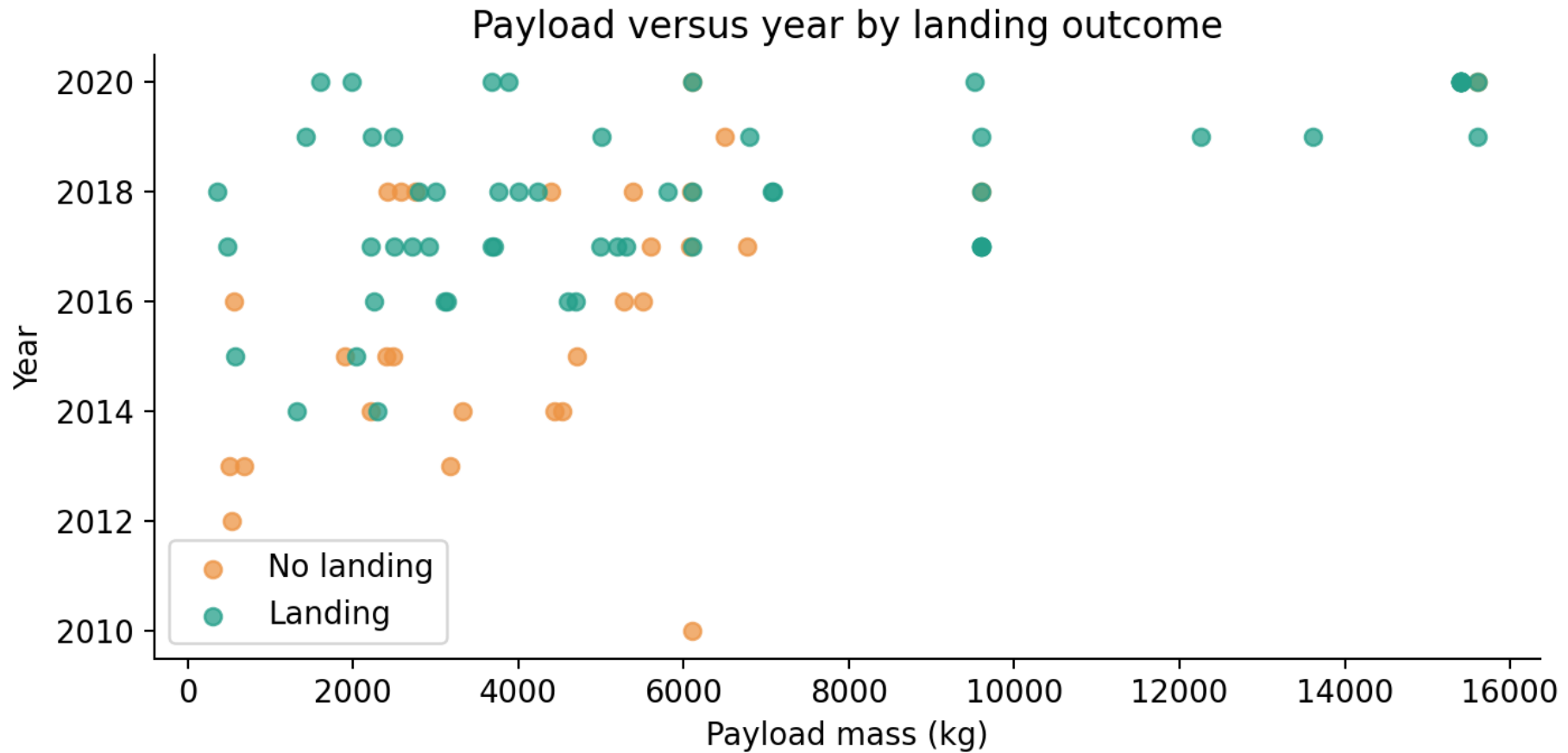
EDA: yearly landing trend

Early years have very few records



EDA: payload, time and outcome

A joint view helps separate year trends from mission mix



SQL results: site and payload

101-row historical table

Launch site	Launches	Landings	Rate
CCAFS SLC-40	34	25	73.5%
CCAFS LC-40	26	6	23.1%
KSC LC-39A	25	20	80.0%
VAFB SLC-4E	16	10	62.5%

Payload mass: minimum 0 kg; maximum 15,600 kg; mean 6,138 kg.

SQL results: orbit ranking

Top eight orbit labels by number of launches

Orbit	Launches
GTO	30
LEO (ISS)	26
LEO	25
Polar LEO	8
SSO	6
MEO	3
HEO	2
Sub-orbital	1

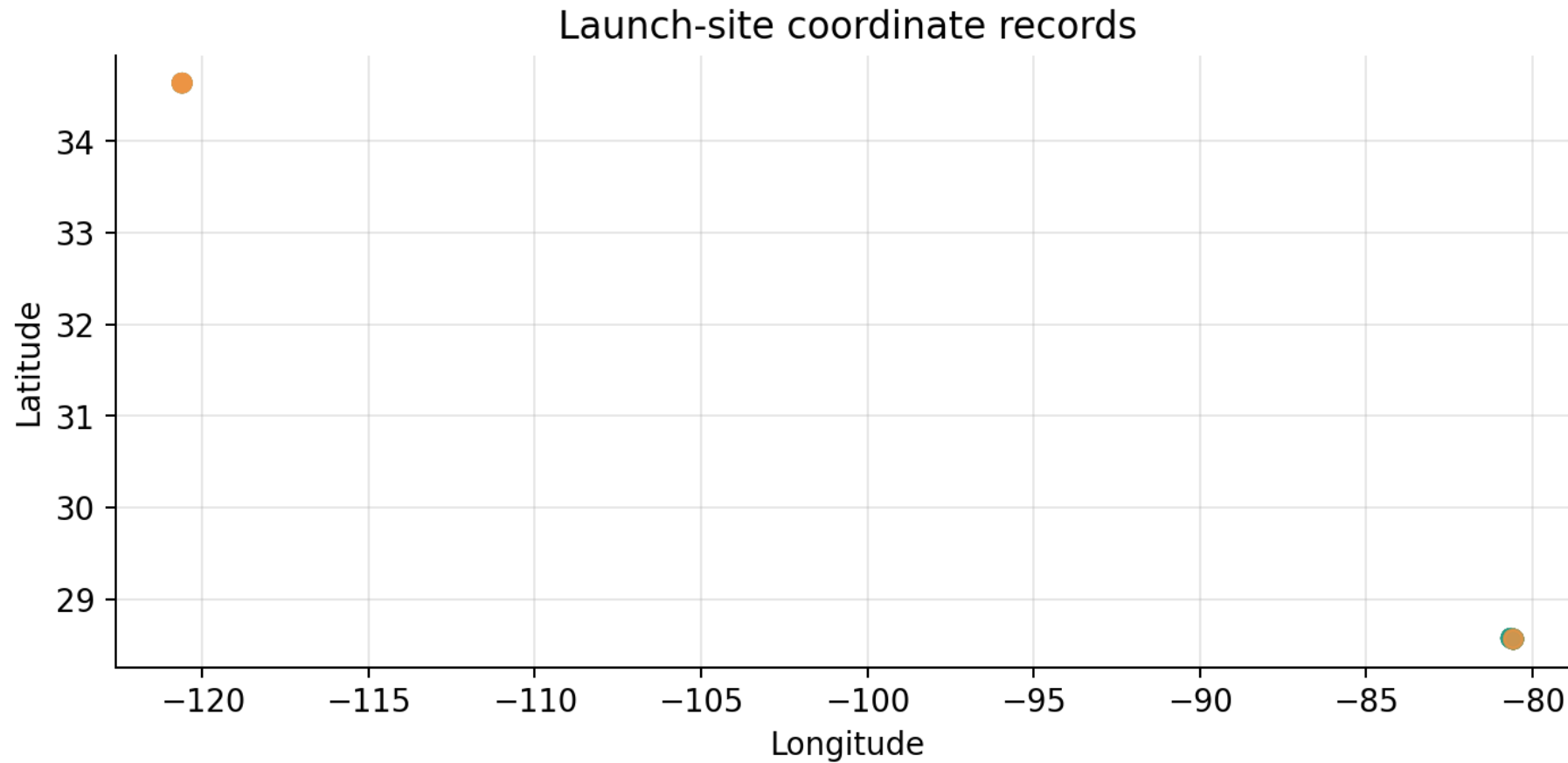
SQL results: time analysis

SQL counts by year, with successful landing records

Year	Launches	Landings
2010	2	0
2012	2	0
2013	3	0
2014	6	0
2015	7	1
2016	8	5
2017	18	14
2018	20	10
2019	11	10
2020	24	21

Folium map: sites and markers

Static preview of historical 56-row coordinate dataset



Folium map: outcomes and proximity

Interactive marker tooltips and five-kilometre circles

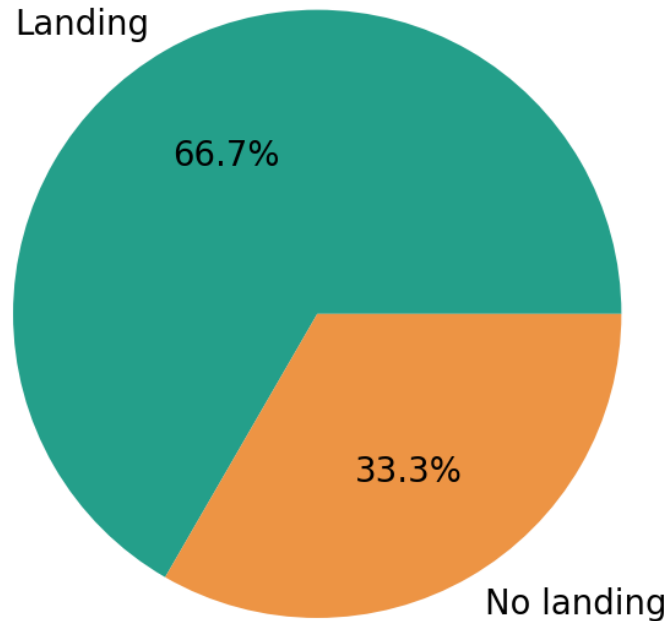
Site	Records	Landing rate
CCAFS LC-40	26	26.9%
CCAFS SLC-40	7	42.9%
KSC LC-39A	13	76.9%
VAFB SLC-4E	10	40.0%

The circle is a visual distance guide. No claim about distance to coastlines, railways or roads is made.

Plotly dashboard: success composition

Historical 90-row model cohort shown for report consistency

Historical dataset outcome share



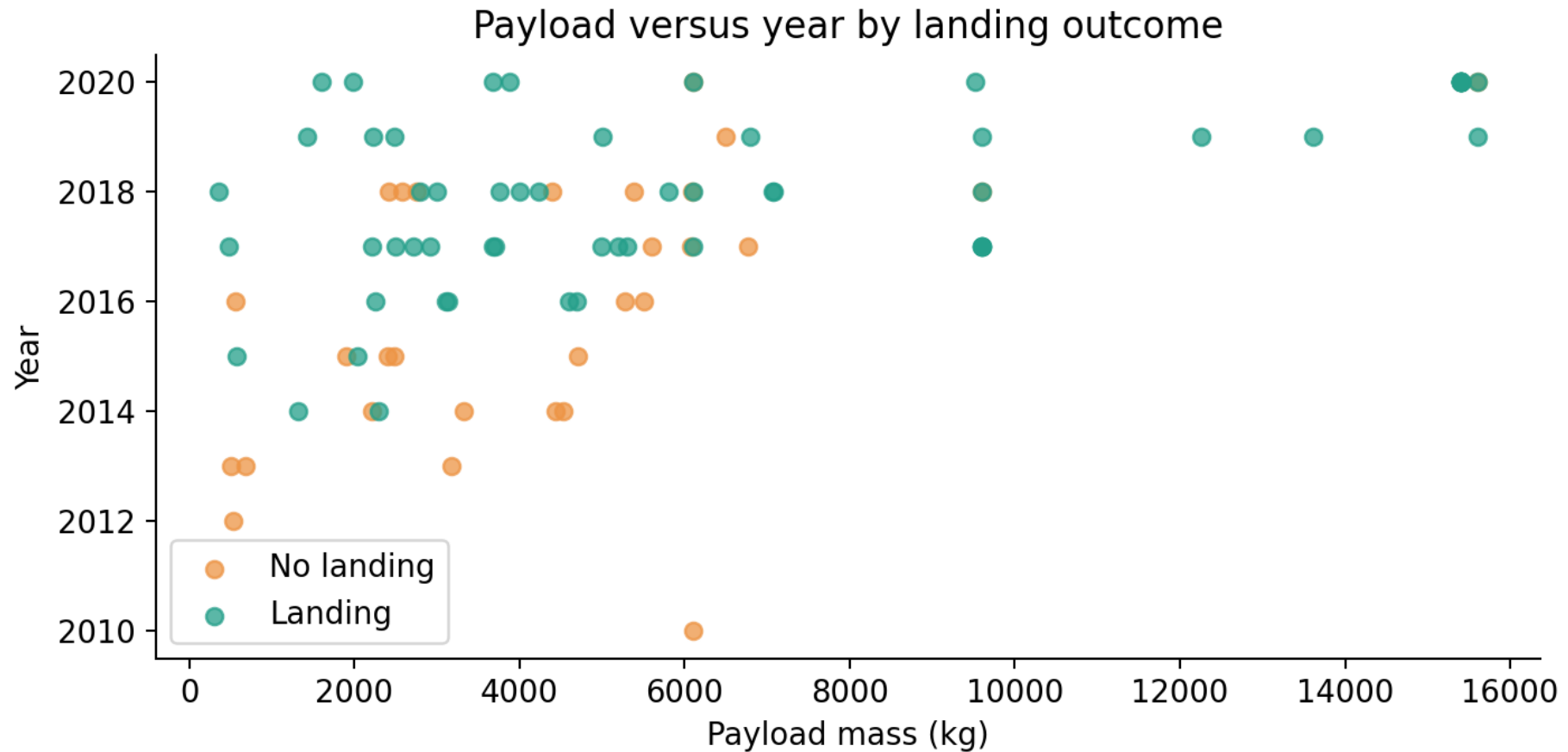
Landing: 60 of 90 (66.7%).

No landing: 30 of 90 (33.3%).

The interactive HTML uses the separate 56-row course dashboard dataset and includes a site pie and payload/outcome scatter.

Plotly dashboard: payload and result

Static preview; inspect interactive scatter in the HTML file



Predictive analysis: classification

Training design and controls

Predictors: payload mass, booster flights, grid fins, reuse, landing legs, block, reuse count, orbit and launch site.

Stratified split: 72 training rows, 18 held-out test rows. Five-fold stratified GridSearchCV selects hyperparameters by training accuracy.

Logistic regression, support vector machine, decision tree and K nearest neighbors are compared. The test set is held out from tuning.

Baseline test accuracy from always predicting landing is $12/18 = 66.7\%$.

Predictive analysis: four models

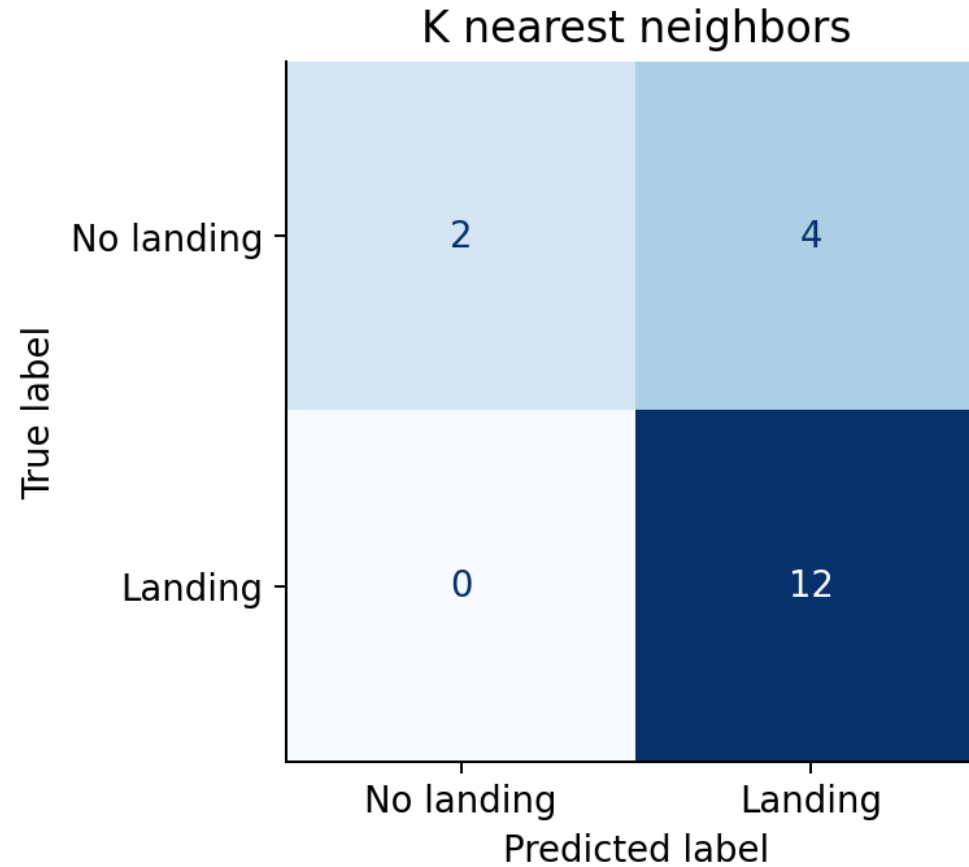
Cross-validation selects the model; test results audit it

Model	CV accuracy	Test accuracy	Test F1
Logistic regression	86.0%	77.8%	0.857
Support vector machine	87.4%	77.8%	0.857
Decision tree	84.6%	77.8%	0.857
K nearest neighbors	87.4%	77.8%	0.857

SVM and KNN tie at 87.4% cross-validation accuracy. The deterministic tie rule selects KNN. All four score 77.8% on this small test split.

Predictive analysis: confusion matrix

Selected KNN model on 18 held-out rows



Conclusions and innovative insights

Three results that affect interpretation

1. Test accuracy rises 11.1 percentage points above the majority-class baseline, yet the selected model misses four of six failed landings. Accuracy alone hides that operationally relevant error.
2. Site differences are sizable in this cohort (60.0% at CCAFS versus 77.3% at KSC), but site and year distributions may differ. These are descriptive comparisons.
3. Orbit rates of 0% or 100% for one-record groups are unstable. Counts must accompany percentages in any decision briefing.

Historical data and an 18-row test set limit generalization. A later cohort would be needed for external validation.

Repository and reproducibility

Project files and data provenance

GitHub repository: <https://github.com/phyonyeinchan/ibm-spacex-capstone>

The public repository contains the executed notebook, Python analysis script, archived IBM CSV inputs, SQL result tables, interactive HTML and this PDF.

Data provenance and exact source URLs are in README.md. Run `python analysis.py` from the project folder to regenerate results.

The notebook records outputs from the historical course datasets. Results should not be interpreted as current SpaceX performance.